

## github

- Create Repo ✓
- write code
- Commit
- Push

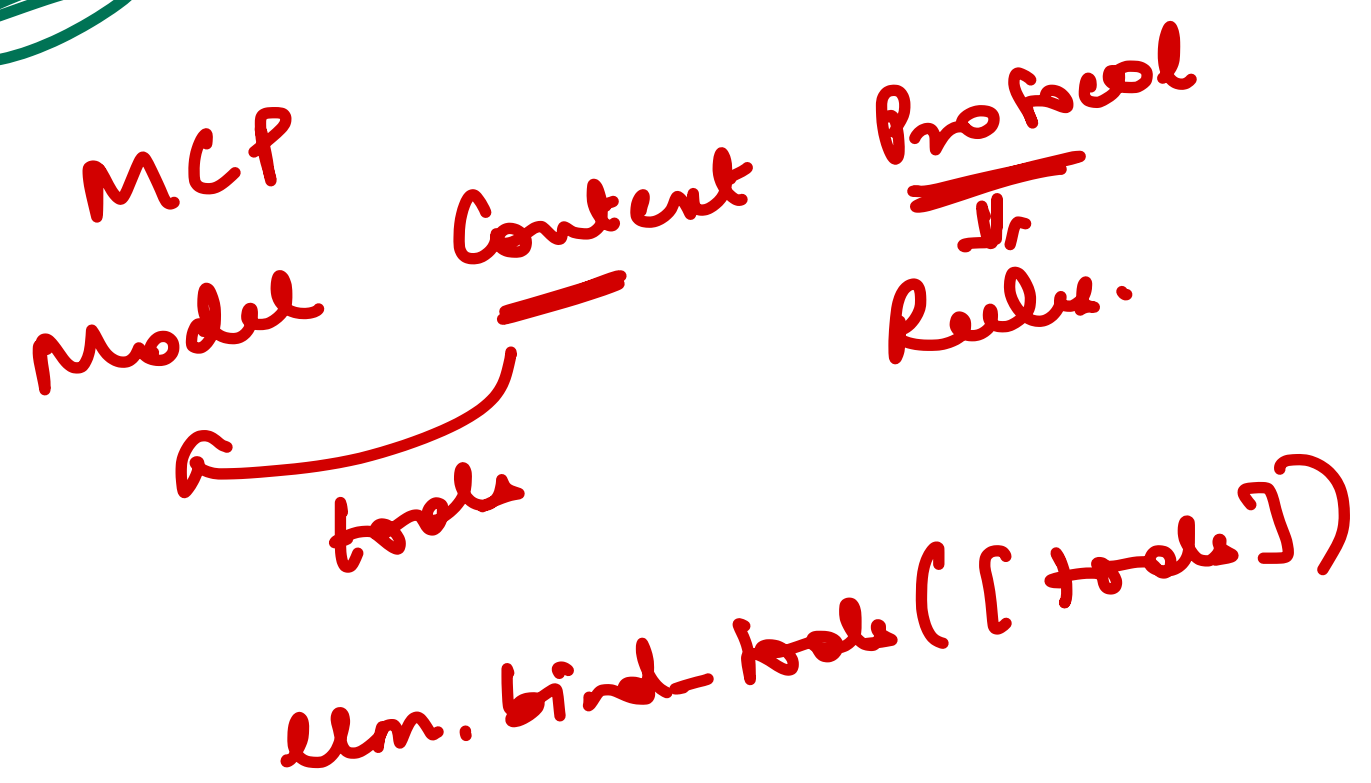
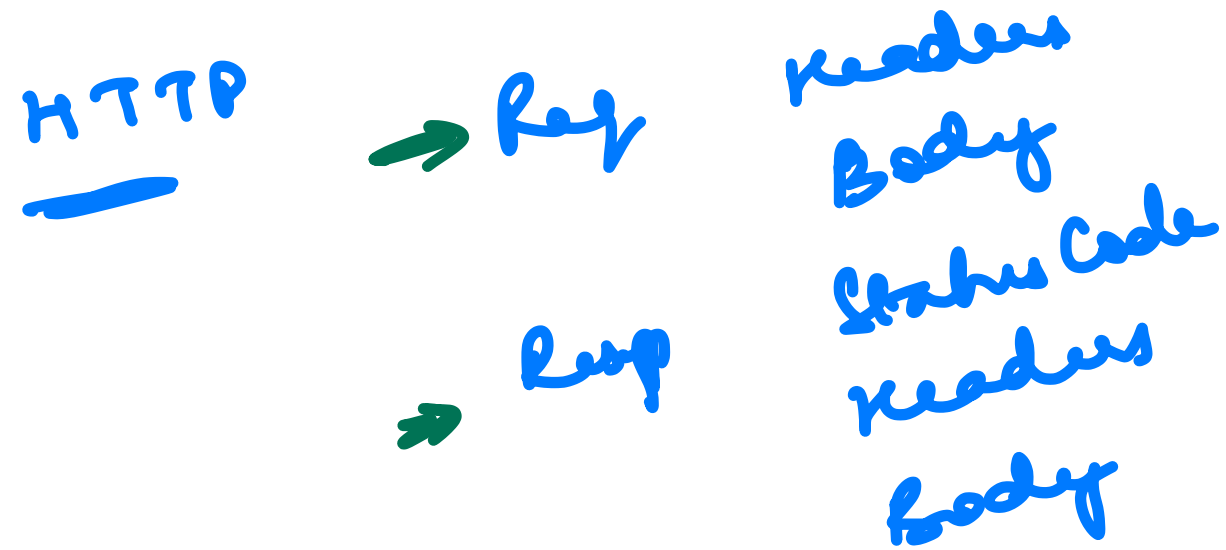
APP → Prompt →

langchain / langgraph

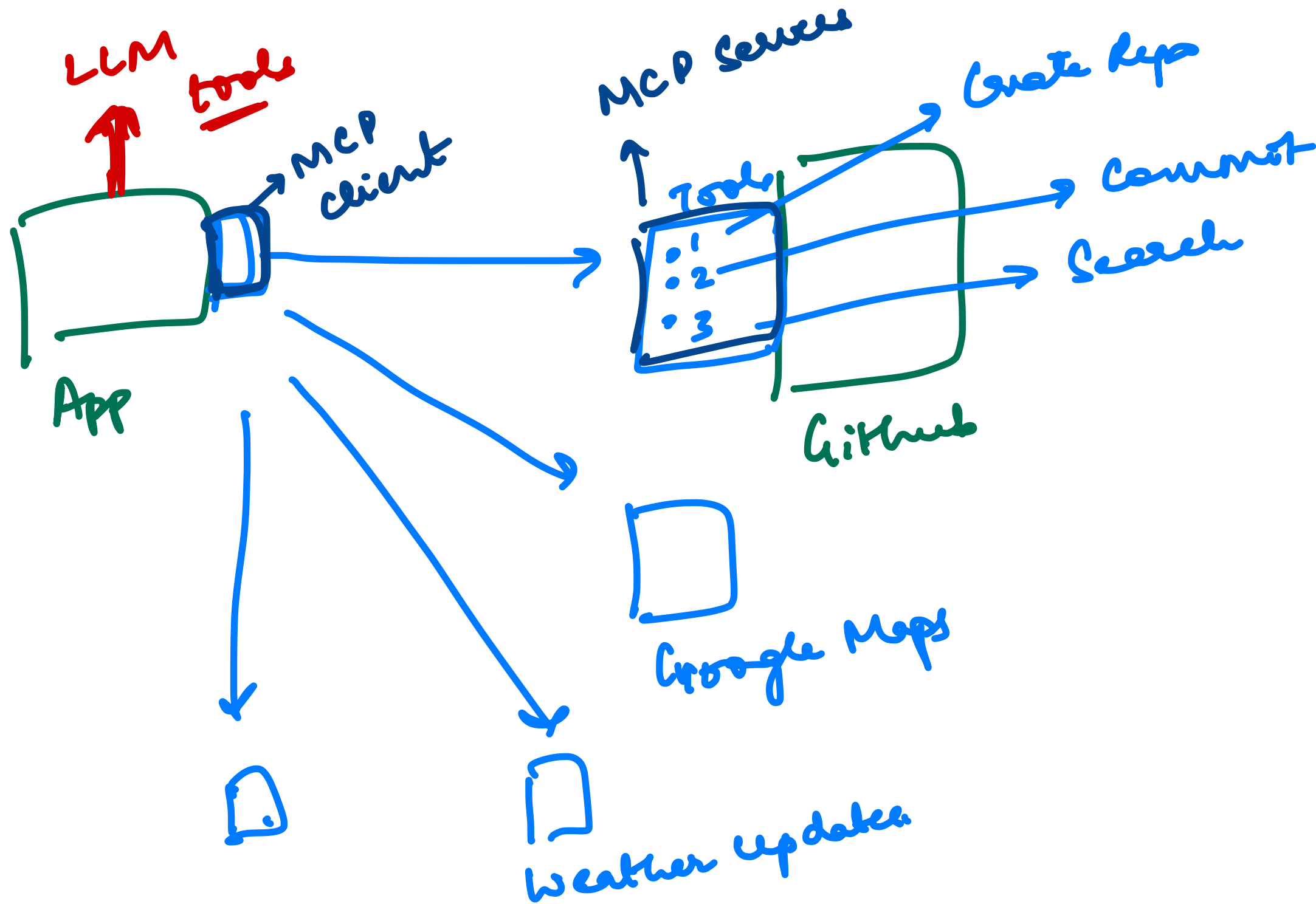
tool → func

{ Create Repo on Github

}  
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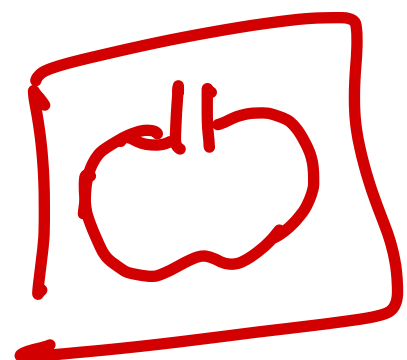
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Image to text  
video to text  
text to Image

[

vectors /  
embeddings

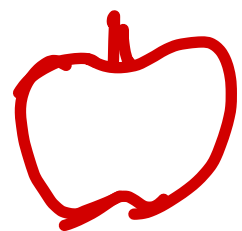
]



"apple"



CLIP



[



]



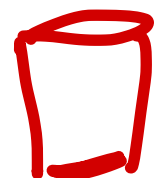
]



"apple"



[



[



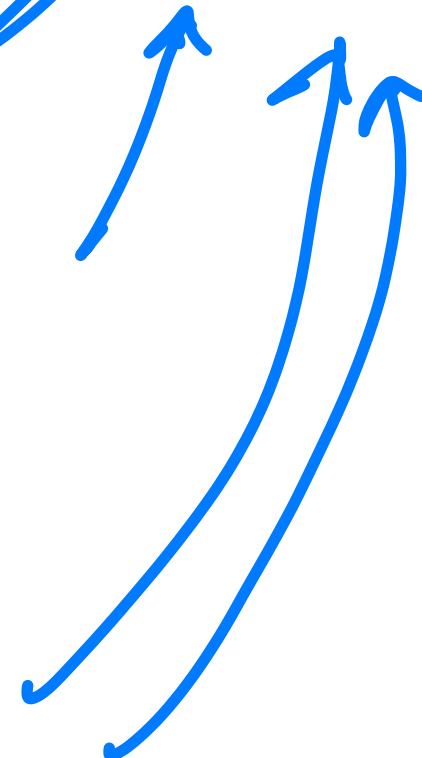
]

]

"glass"



[



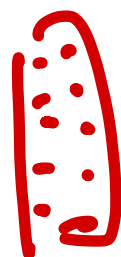
remote



[

]

]



[



]

Image → Text

Cat sat on a mat





v1  $\rightarrow$  Jan 2024  $\rightarrow$  67B  
traditional transformer  $SA + \boxed{MHA}$

v2  $\rightarrow$  236B  $\rightarrow$  MoE  $\rightarrow$  Mixture of Experts  
 $\rightarrow$  MHA  
Multiheaded Latent  
Attention

v3  $\rightarrow$  671B  
Reinforcement Learning

---

R1  $\rightarrow$  Reasoning  
 $\downarrow$   
Jan 2025

## 4 Points

① RL

② MoE

Chain of Thought  $\rightarrow$  Reasoning

③

Distillation  $\rightarrow$  Deepseek  
local

④



allama → Run on local

---

Model

NN

Experte

NN<sub>1</sub>

NN<sub>2</sub>

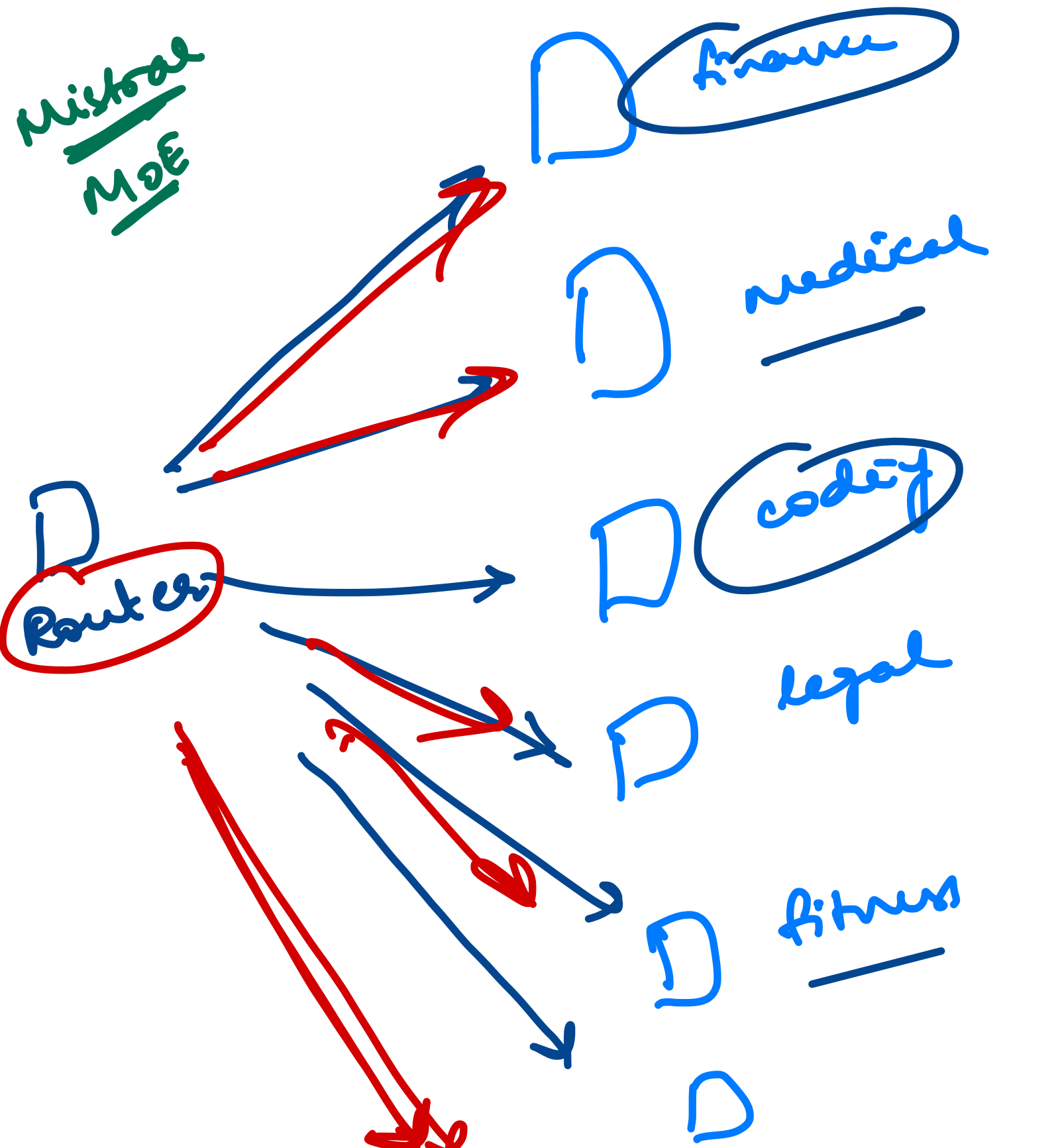
NN<sub>3</sub>

NN<sub>4</sub>

Mistral  
MoE

Specialised  
module

domain /  
knowledge



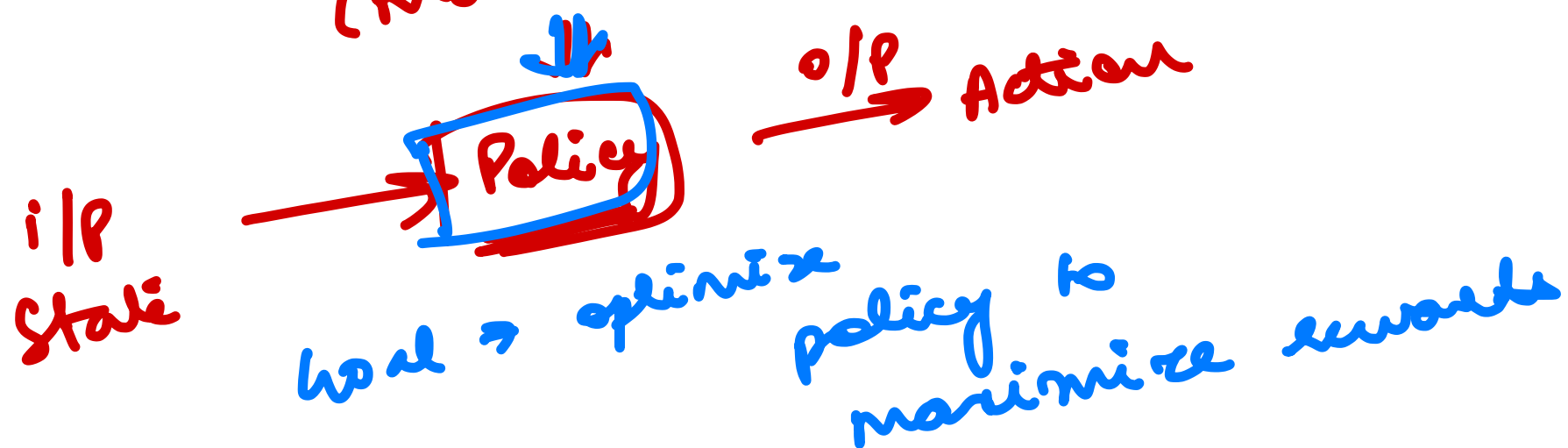
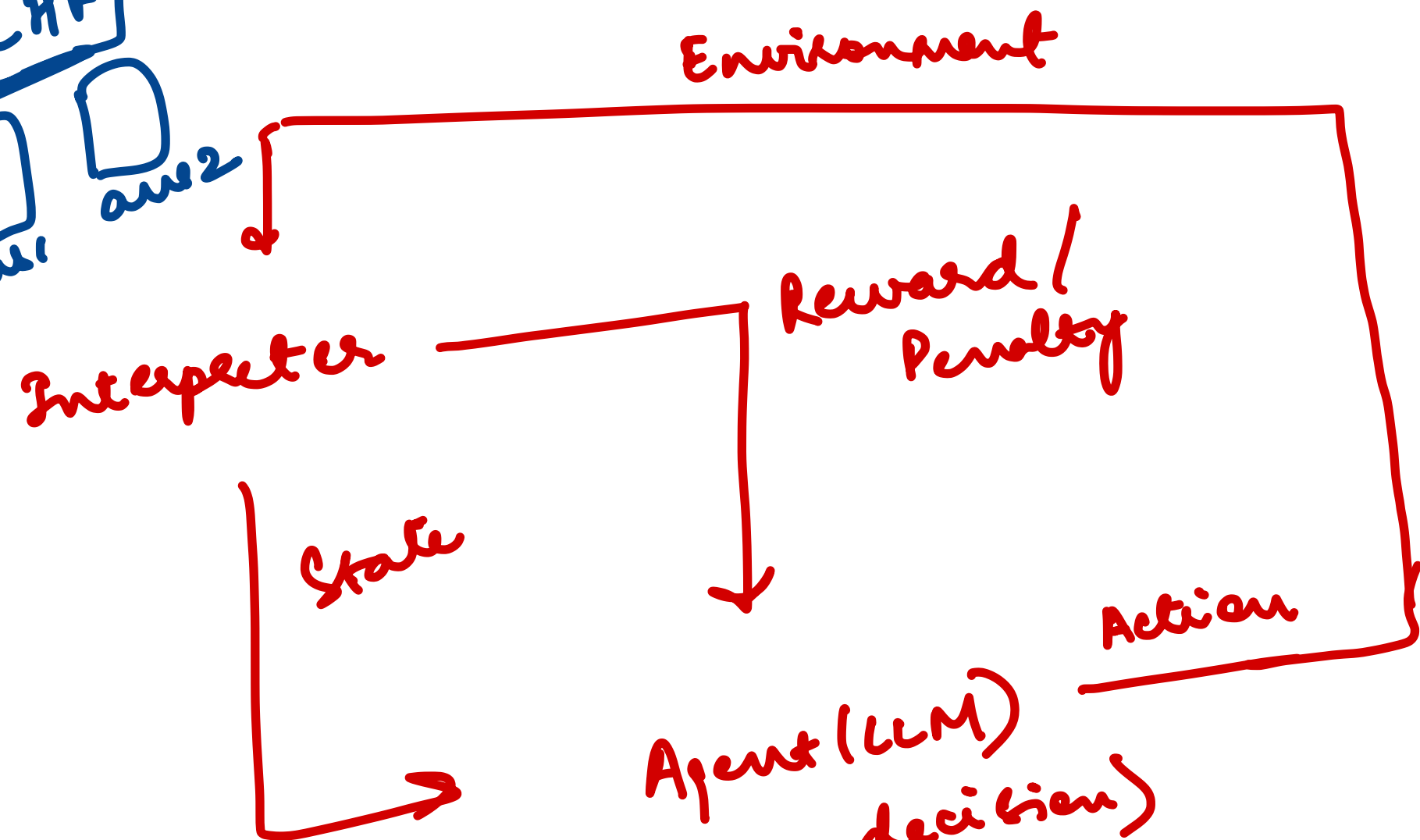
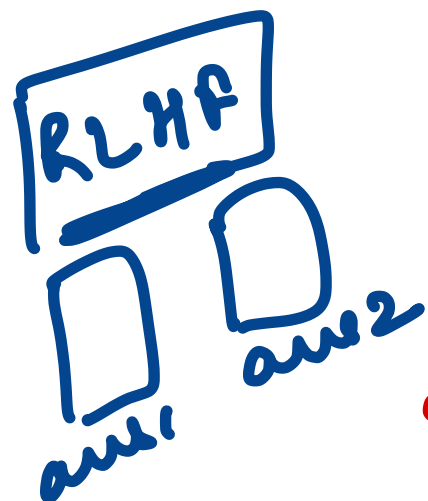
Shared Experts → General  
Experts

GAN  
Mode collapse  
generating image  
came image

Activated Parameters

Total = 671B

"37B activated parameters"



Model

↓ Fine Tuned  
Instructions

Instruction Tuned Model

↓ Fine Tuned

RL → ✓

# Box Model - Transformer Architecture

Five Turn Instructions

Chat Foundation Models

v3, GPT 3.5

Reasoning Five Turn

Reasoning Models

Car → Speed, Distance  
time time to reach?

Think out loud



$$S = \frac{D}{T}$$

$$D = S \times T$$

time + \_\_\_\_\_

time  
Dist

House  $\rightarrow$  5th area.  $\rightarrow$  W, L

Carpet.  $\rightarrow$  Carpet do I need.

$$\frac{W \times L - \text{Carpet area.}}{\text{Carpet area}}$$

Code / Maths / logical

Explain step by step

Chain of Thought

→ COT

→ Step 1 →  $L \times W$

→ Step 2 → Carpet

→ Step 3 → Subtract

Aaha → Making mistake

↓  
Reevaluate  
↓  
better ans



Mixture of Experts  $\Rightarrow$  Computation is faster. more efficient

Reinforcement Learning  $\Rightarrow$  Self Improvement  
Model guides itself

Chain of Thought  $\Rightarrow$  More transparency  
Self Evaluation  
Performance is better

---

Distillation  $\Rightarrow$  More available /  
accessible

Problem → Models are too big

Distillation Efficiently

Teacher Model

Big, Powerful Model

↓  
General

→ Better, Not Bigger

Student Model

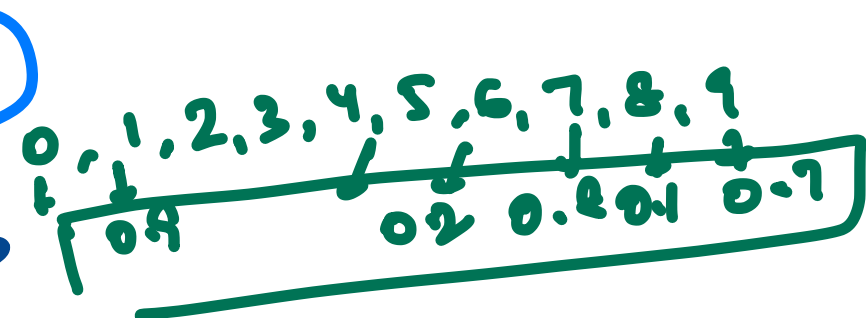
Smaller Model  
Performing Better

⇓  
Specialized  
Small

Teacher

[1]

→



[2]

→

2

[3]

→

3

[4]

→

4

⋮

MNIST Dataset

Student

[1]

→

2 → 2

3 → 3

⋮

,

⋮

⋮

⋮

→ Pro Labels

2 Methods

① Soft labels / Soft targets

Teacher

8000

1

o/p

0 0

0 1

0 2

0 3

0 4

0 5

6

7

.

1

Sum = 1  
(Probab)

0.8

0.004

0.03

0.1

Student

lesser  
params

2B

1 → ———

2 →

3 →

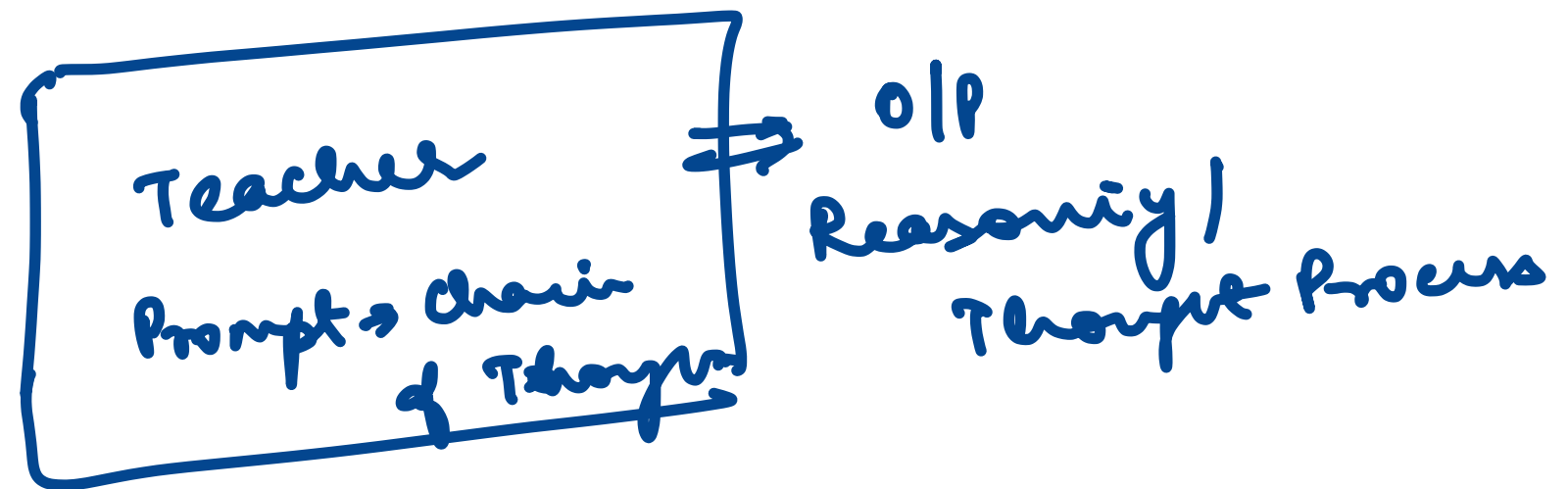
4 →

↓

Specializing

Fine Tuning  
Student  
using the o/p  
prob abs from  
our teacher.

# Reasoning / Thought Process



O/P +  
Thought Process

AT Student

Specialized at Reasoning

Distillation Step by Step

Efficient → less computation

Smaller → Accuracy ✓ → Specialized domains

✓ Huge, Powerful Model  
- billions of parameters

(General)

→ Generating Reasoning behind ans

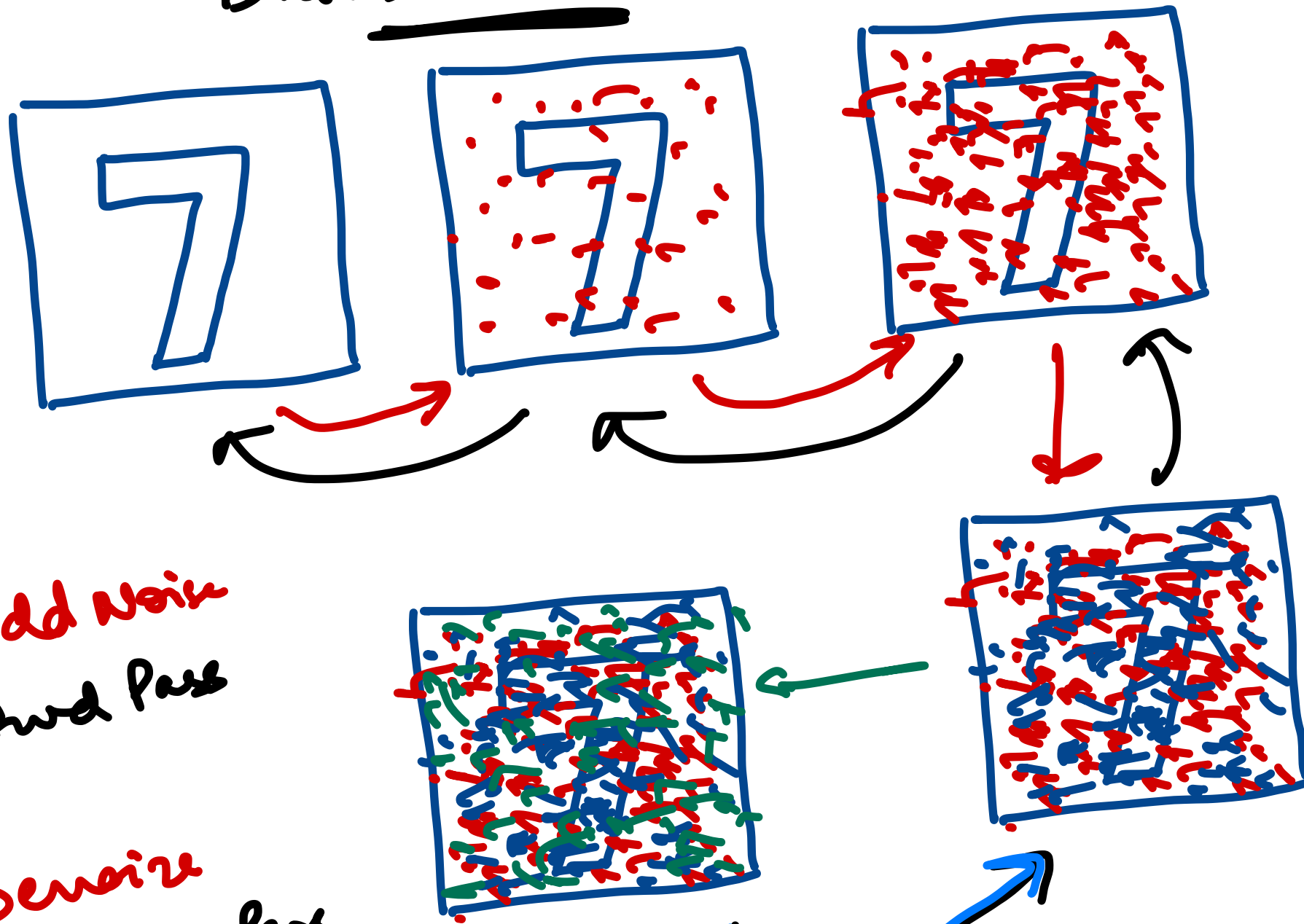
→

| IP | OP | Reasoning |
|----|----|-----------|
|    |    | ✓         |
|    |    | ✓         |
|    |    | ✓         |
|    |    | ✓         |

⇓  
Student Model → smaller  
less no. of params



# Diffusion Model

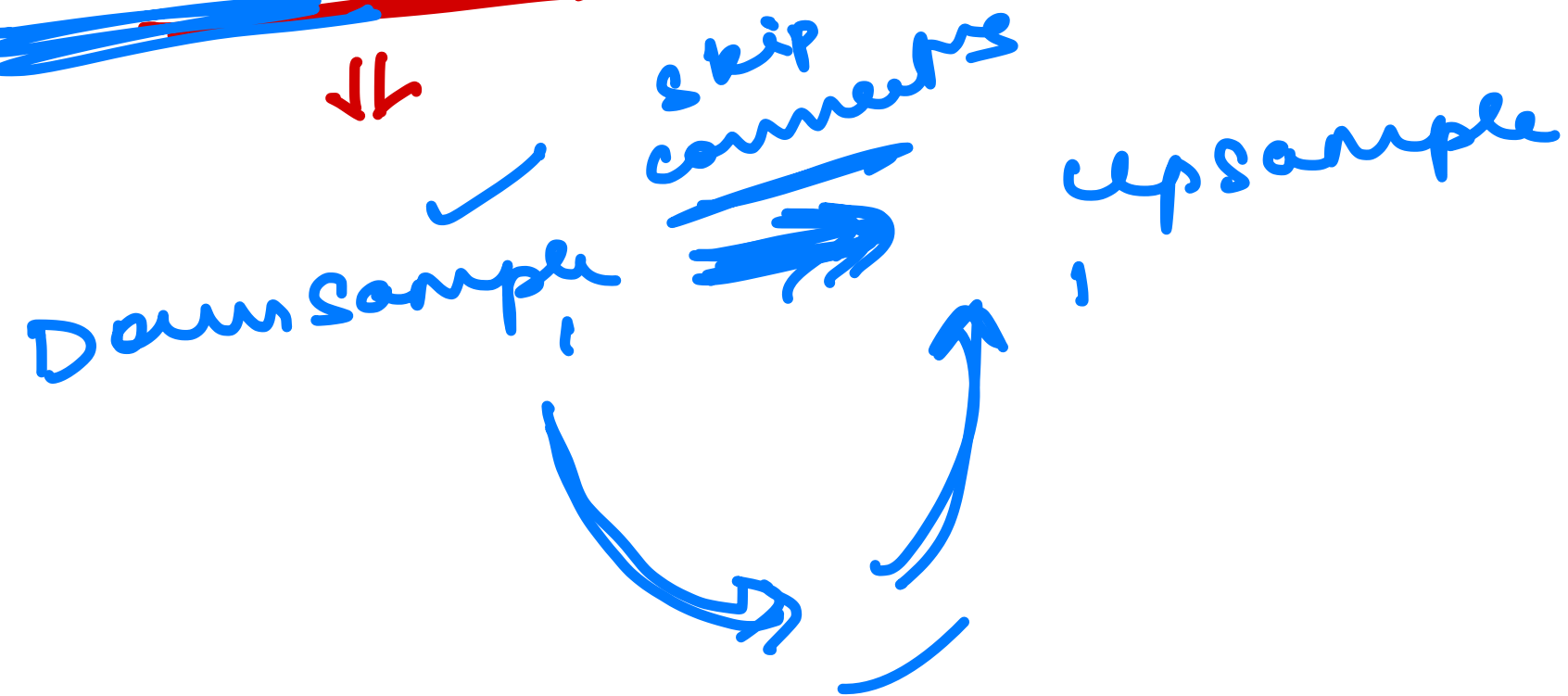


① Add Noise  
→ Add Pass

② Denoise  
→ Reverse Pass

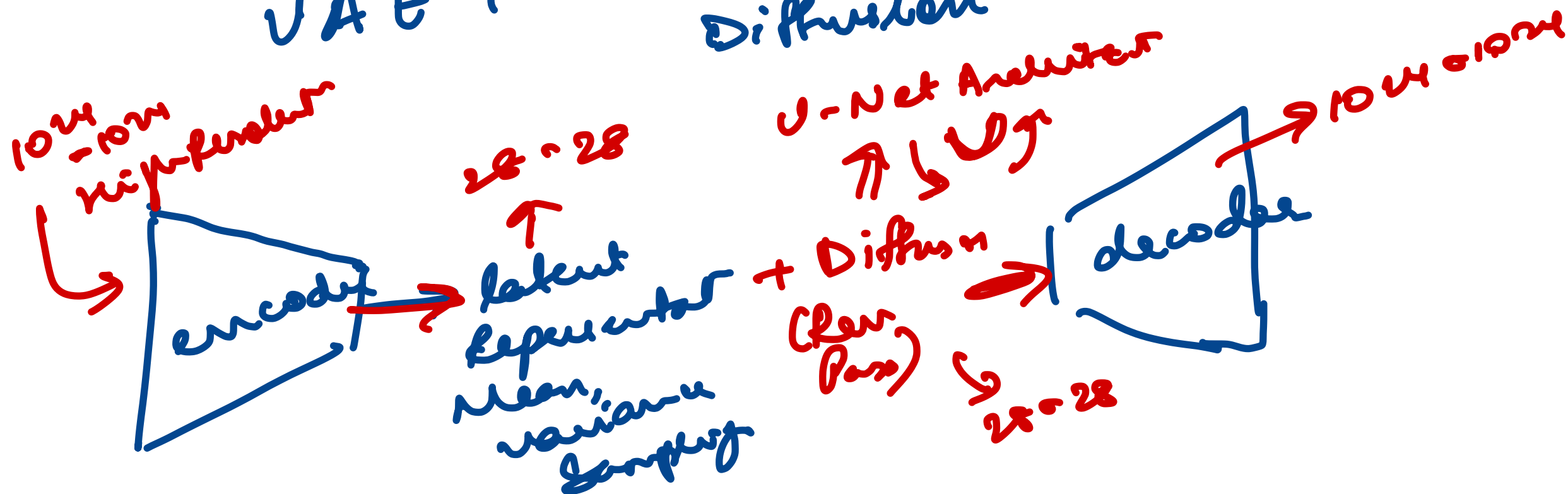
Train → Denoise in Steps  
to generate images

# V-Net Architecture



## Stable Diffusion

VAE + Standard Diffusion





Text to Image

↓  
Embedding ↔ Embedding.

OpenAI Embedding

Textual Meaning of Sentence

Tokenize

↓  
Encode

↓  
Embed

word2vec

NN

Cats and dogs are playing together.

visual meaning →

# Text to Image

①

I/P = Text

→

Embed

using

CLIP

→ captured meaningful image representation

②

Random Noise

↳ Denoise

→

Image

⌋  
⌋  
⌋

+

Extra Info

Embedding of

Text

⇒

K, V  
vectors

Cross Attention

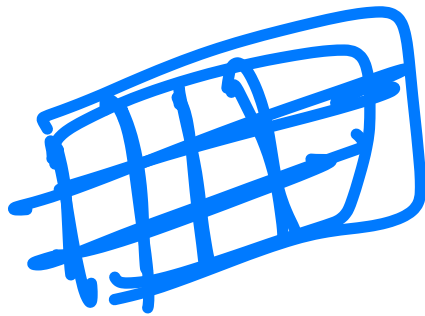
# CLIP

Text, Image

→ Tokenize → Patches / Part

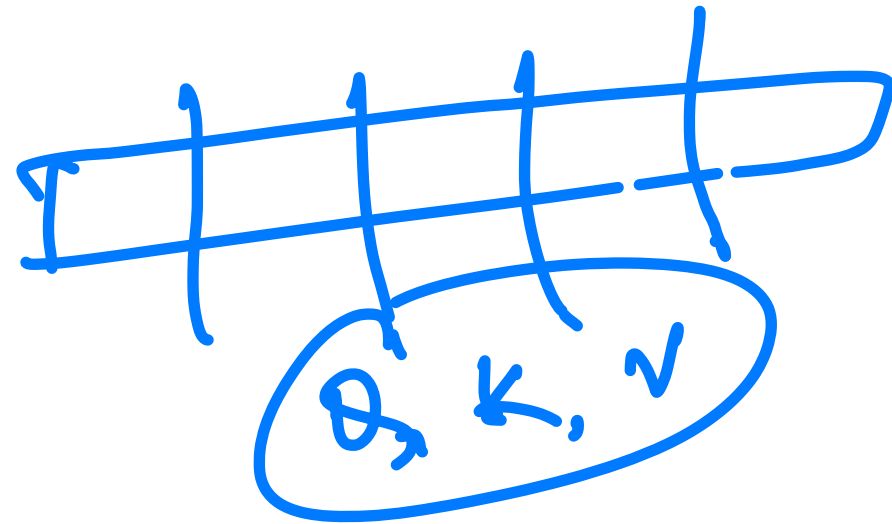
↪ Encoder

Image



↓  
Embedding  
Representer

Decoder



(Images) → Meaning  
Image Classification

Recognition  
Encoder

Generation  
Dall-e  
Midjourney  
Decoder

(BERT)

GAN

VAE

Diffusion Models

Stable Diffusion - LDM

Latent Diffusion Models

(VAE + DM)

Transformer Architecture } → generate images

Transformer + Diffusion  
↓  
CLIP  
↓  
generate images

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